

Nicholas Kevin Hu

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EDUCATION

Rice University, Houston, TX

May 2027

BS in Electrical and Computer Engineering, Minor in Data Science

GPA: 3.7/4.0

Relevant Coursework: Computational Thinking and Data Structures, Fundamentals of Computer Engineering, Fundamentals of Electrical Engineering, Introduction to Physical Electronics, Linear Algebra, and Multivariable Calculus

SKILLS

- **Languages:** Native proficiency in English and Chinese, Proficient in Spanish
- **Relevant Skills:** Python, Java, MATLAB, Simulink, Raspberry PI, Arduino, GitLab, Data Science, Machine Learning, CITI Certified, PCB Design, Microsoft Office Suite (Word, Excel, PowerPoint), Object-Oriented Programming (OOP)

WORK EXPERIENCE

Rice University Electrical and Computer Engineering, Student Researcher, Houston, TX

November 2024 - Present

- Collaborated with Baylor College of Medicine researchers to develop machine learning algorithms for predicting patient mood and depression levels through biomarkers, enhancing post-amputation mental health care and interventions
- Obtained CITI certification to ensure ethical research practices and compliance with institutional guidelines for handling sensitive patient data.

Rockwell Automation, Control and Firmware Engineering Intern, Milwaukee, WI

June 2024 - August 2024

- Conducted efficiency and starting torque testing using the Test Automation Framework; optimized performance and assisted with data collection for drive and firmware development by running over 200 tests
- Gained core knowledge of Lean Six Sigma principles and methodologies for process improvement and operational efficiency, utilizing Lean tools such as 5S, Kaizen, and DMAIC (Define, Measure, Analyze, Improve, Control) framework

Lubar Innovation Center, Center and Lab Assistant, Milwaukee, WI

June 2023 - August 2023

- Collaborated closely with researchers to operate and maintain machinery, demonstrating strong teamwork and communication skills in a technical environment
- Provided comprehensive instruction to students on the correct procedures for handling machinery, emphasizing safety and operational efficiency

PROJECTS

MATLAB/Simulink Git Comparison Tool – Rockwell Automation

August 2024

- Leveraged GitLab and MATLAB scripting to automate change detection and reporting, cutting manual intervention by 40% and increasing productivity
- Successfully integrated the tool into existing version control workflows, leading to adoption by 10+ senior engineers across multiple teams

A Machine Vision-based Assistive Device for the Visually Impaired - USM

April 2023

- Proposed and created a novel machine vision-based assistive device that uses convolutional neural networks for object and street sign recognition with a lightweight, portable design powered by Raspberry Pi 4B

LEADERSHIP EXPERIENCE

Institute of Electrical and Electronics Engineers Club – Sophomore Representative

2023 – Present

- Act as a liaison between sophomore students and club leadership to address member needs and concerns
- Assist in organizing and promoting over 10 technical workshops, guest lectures, and industry networking events

OwlSat (Rice Cube Satellite Club)– Hardware Co-Team Lead

2024 – Present

- Coordinate cross-functional collaboration among mechanical, electrical, and software teams comprising over 20 members to ensure seamless integration of satellite hardware systems
- Optimize and reduce redundancy in hardware designs by over 7% to meet size, weight, and power constraints of CubeSat

Global Research and Consulting Group Club – Team Lead

2023 – Present

- Lead a team of 7 to develop a comprehensive content marketing strategy, utilizing case studies, social media, and other marketing tools for the Greater Killeen Chamber of Commerce's transportation initiative

Rice Center for Career Development – Peer Career Advisor

2024 – Present

- Deliver career development workshops and events on topics such as networking, personal branding, and internship searching, inviting over 13 alumni and reaching a student audience of over 100 students to date